

# HOJEONG LEE

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## EDUCATION

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### University of Wisconsin–Madison

Ph.D. in Computer Sciences

Advised by Prof. Suman Banerjee

Sep 2025 - Present

Madison, WI, USA

### Korea University

M.S. in Computer Science and Engineering

Advised by Prof. Hyogon Kim

Mar 2022 - Aug 2025

Seoul, Korea

### Korea University

B.E. in Computer Science and Engineering

Mar 2016 - Feb 2022

Seoul, Korea

## RESEARCH EXPERIENCE

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### WiNGS Lab, University of Wisconsin–Madison

Graduate Researcher

Sep 2025 – Present

Madison, WI, USA

### Wireless Data Communications Lab, Korea University

Graduate Researcher

Mar 2022 – Aug 2025

Seoul, Korea

### Internet Systems Lab, University of Colorado Boulder

Visiting Scholar

Collaborated with Prof. Sangtae Ha and Prof. Seyeon Kim

Fully funded by Korea University

Mar 2024 – Mar 2025

Boulder, CO, USA

### Carnegie Mellon University

Visiting Student

Selected through a merit-based national competition

Fully funded by the Korean Government

Aug 2022 – Feb 2023

Pittsburgh, PA, USA

### Wireless Data Communications Lab, Korea University

Undergraduate Research Intern

Jun 2021 – Feb 2022

Seoul, Korea

## RESEARCH INTERESTS

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### On-device AI & Mobile Systems

Efficient AI inference on mobile devices [P1, P3, P4]

### Networked Multimedia Systems

Real-time streaming systems for media applications [14, P2]

## WORK IN PROGRESS

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P4. **First author**, Speculative decoding.

P3. **First author**, On-device LLM inference.

P2. **Co-author**, Online 3D Gaussian avatar refinement.

P1. **First author**, Efficient speech foundation models.

## PUBLICATIONS

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14. **Hojeong Lee**, Yu Hong Kim, Sangwoo Ryu, James Won-Ki Hong, Sangtae Ha, and Seyeon Kim. DeltaStream: 2D-Inferred Delta Encoding for Live Volumetric Video Streaming. ACM MobiSys, Anaheim, California, USA, 2025.

13. **Hojeong Lee**, Seungmo Kang, and Hyogon Kim. Causality-sensitive scheduling to reduce latency in vehicle-to-vehicle interactions. *Sensors*, 24 (22), 2024.
12. Seungmo Kang, **Hojeong Lee**, and Hyogon Kim. Mitigating Latency Inflation in V2C Transactions Using Periodic Sidelink Communication. *IEEE VNC*, Kobe, Japan, 2024.
11. **Hojeong Lee**, Chanwoo Kim, Eugene Yang, and Hyogon Kim. Distributed Joint Congestion Control for V2X Using Multi-agent Reinforcement Learning. *IEEE ICMLCN*, Stockholm, Sweden, 2024.
10. **Hojeong Lee** and Hyogon Kim. Improving One-Shot Transmission in NR Sidelink Resource Allocation for V2X Communication. *arXiv preprint arXiv:2312.15914*, 2023.
9. **Hojeong Lee** and Hyogon Kim. Rethinking Transmit Power Control for SAE J3161/1 Congestion Control Algorithm. *IEEE VTC2023-Fall*, Hong Kong, 2023.
8. Yeomyung Yoon, **Hojeong Lee**, and Hyogon Kim. Deep reinforcement learning-based dual-mode congestion control for cellular V2X environments. *Electronics Letters*, 59 (20), 2023.
7. Kyeongnam Park, **Hojeong Lee**, and Hyogon Kim. Speed-Aware V2X Congestion Control. *IEEE VTC2023-Fall*, Hong Kong, 2023.
6. Kyeongnam Park, Kyungha Kim, Hyungjoon Shin, **Hojeong Lee**, and Hyogon Kim. Strategically Positioning On-Board PEPs in LEO-based NTN for TCP Throughput Improvement. *IEEE VTC2023-Fall*, Hong Kong, 2023.
5. Hyeonji Seon, **Hojeong Lee**, and Hyogon Kim. Predicting CAM generation times through machine learning for cellular V2X communication. *ICT Express*, 9 (5), 2023.
4. Joseph Konan, Ojas Bhargave, Shikhar Agnihotri, **Hojeong Lee**, Ankit Shah, Shuo Han, Yunyang Zeng, Amanda Shu, Haohui Liu, Xuankai Chang, Hamza Khalid, Minseon Gwak, Kawon Lee, Minjeong Kim, and Bhiksha Raj. Improving Perceptual Quality, Intelligibility, and Acoustics on VoIP Platforms. *arXiv preprint arXiv:2303.09048*, 2023.
3. Hyeonji Seon, **Hojeong Lee**, and Hyogon Kim. Packet Delivery Impact of Predictive Resource Allocation for Quasi-Periodic Cellular V2X Communication. *IEEE VTC2023-Spring*, Florence, Italy, 2023.
2. Jonghwan Na, **Hojeong Lee**, and Hyogon Kim. Inferring Human Driver Intent in Partial Deployment of Connected Autonomous Vehicles: the Lane Change Case. *IEEE VTC2023-Spring*, Florence, Italy, 2023.
1. Hyeonji Seon, **Hojeong Lee**, and Hyogon Kim. Prediction of vehicle dynamics-based aperiodic message generation times in cellular V2X communication. *Annual Conference of KIPS*, Seoul, Korea, 2022. **Outstanding Paper Award**

## AWARDS & GRANTS

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### Summer Graduate Fellowship Award, University of Wisconsin–Madison

Department of Computer Sciences, Summer 2026

### Scholarships for Internationalization, Korea University

Visiting scholar funding, University of Colorado Boulder, Fall 2024

### Conference Student Travel Grants

IEEE ICMLCN 2024, IEEE VTC2023-Fall

### Department Merit-Based Scholarship, Korea University

Fall 2021, Spring 2021, Fall 2020

### Korean Government Scholarship

Fall 2021, Spring 2021, Fall 2020, Fall 2016

## TEACHING EXPERIENCE

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### Teaching Assistant, University of Wisconsin–Madison

Introduction to Computer Networks, Spring 2026

Foundations of Mobile Systems and Applications, Fall 2025

## Teaching Assistant, Korea University

Internet Protocol, Spring 2023

Computer Network, Spring 2022

## EXTRACURRICULAR ACTIVITIES

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### Vice President, Korean Students and Scholars Association, University of Wisconsin–Madison

Jun 2026 – Present

Support operations, financial management, and community-building initiatives

### Volunteer, ACM IMC 2025

Oct 2025

### Coach, NAVER Corporation

Jan 2021 – Aug 2021

Boost Course PY4E (Python programming) and Harvard CS50

Selected as an excellent coach in PY4E

### Software Education Volunteer, Korea University

Aug 2020 – Dec 2020

Software camp instructor and mentor for middle and high school students

Software-related educational video production and Arduino project coaching

### Military Service, Republic of Korea Army

Jul 2018 – Mar 2020

Division Commander's Award, 3rd Place, combat mission-oriented physical training competition

### Overwatch Professional Gamer, Team LW Red

Jan 2017 – Dec 2017

Runner-up, Overwatch National University Tournament Season 2

Sep 2017

Winner, Overwatch APEX Challengers Season 4

Jul 2017